

AI 301 · UNIT 3

# Fix It

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You drive, AI operates.

# Welcome to Unit 3: Fix It

- 1 **Turn on** your camera
- 2 **Rename** yourself on Zoom if needed
- 3 **Answer** tonight's question in the chat

TONIGHT'S QUESTION • IN THE CHAT

**What was your biggest takeaway from last class?**

## OUR COMMUNITY

# As a community, we will...

- \* **Be present**, physically and mentally
- \* **Actively build** an inclusive community and space
- \* **Practice** curiosity and adopt a learning mindset
- \* **Ask** for help when we need it
- \* **Keep** our cameras on

CODEPATH\*ORG | WEEK 1 + 2 ASSIGNMENTS

# You have more time to finish

Amazing job to everyone who submitted on time. We want to bring everyone to the finish line, so Week 1 + 2 assignments stay open through Sunday, Oct 4.

EXTENDED DEADLINE

# Oct 4

Sunday

## HAVEN'T SUBMITTED YET

### Get your work in by Oct 4.

Submit your Week 1 and Week 2 assignments by Sunday, Oct 4.

## ALREADY SUBMITTED

### You can resubmit for a higher score.

AI 301: You can resubmit for a higher score, until the last day of courses this Fall.

## STUCK? WE'VE GOT YOU

### Current support Available now

Help channels on Slack, plus Study Hall sessions.

### Email a Tech Coach

**Starting today, 5pm PT**

Get 1:1 help at [support@codepath.org](mailto:support@codepath.org), in addition to Slack.

### After-session support 7–8pm PT

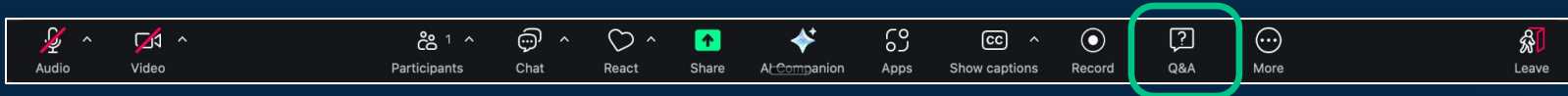
Stay after session for help finishing your assignments so you can submit.

[Zoom link](#)

## ASK ANYTIME

# Add your questions to the Q&A

- \* Our Tech Fellow team adds answers throughout the lesson
- \* Chat questions may be missed or redirected to the Q&A



Q&A, on the right

# Your Feedback



## Glow

- + Shoutout Spencer!  
Lecture was awesome
- + Improved breakout instructions



## Grow

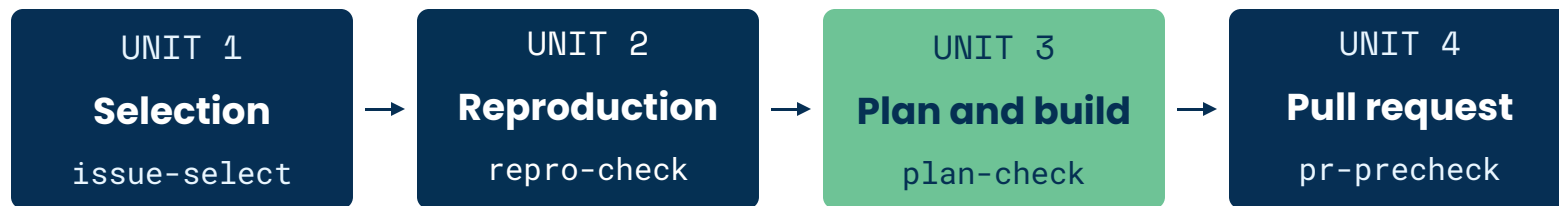
- Spend less time on admin and more on lecture

# Agenda

|    |                 |                                         |        |
|----|-----------------|-----------------------------------------|--------|
| 01 | <b>Intro</b>    | continuing from proof to plan           | 10 min |
| 02 | <b>Lecture</b>  | plan and build                          | 45 min |
| 03 | <b>Break</b>    |                                         | 5 min  |
| 04 | <b>Activity</b> | Teach Claude to Grade a Plan, in groups | 50 min |
| 05 | <b>Wrap-up</b>  | what's due, where this is headed        | 10 min |

# Where this session sits

we're in 🌂 the sandbox! 🌂



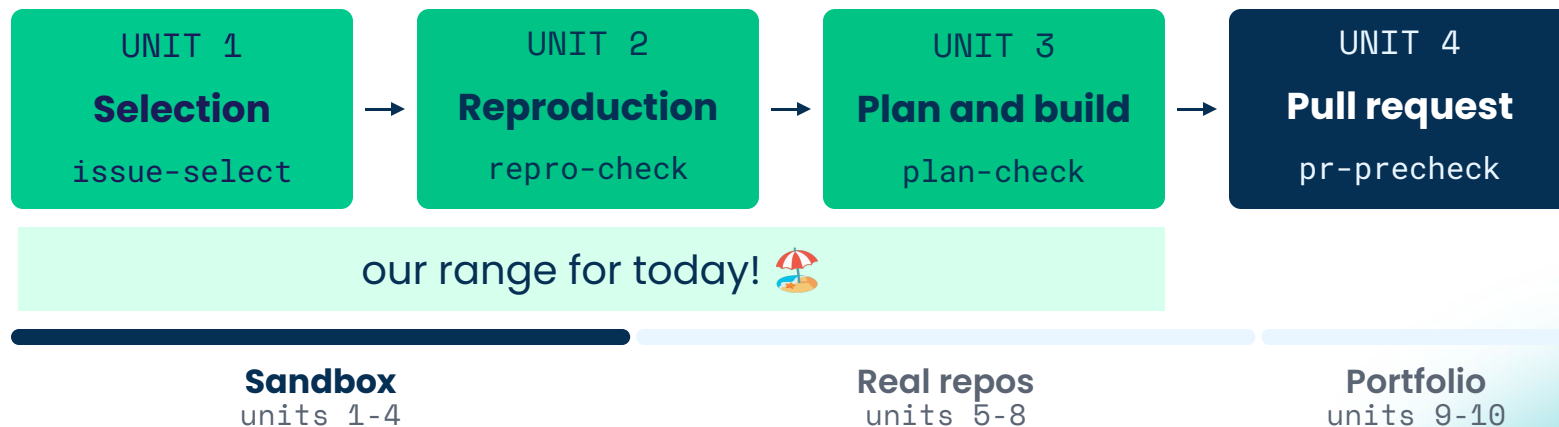
and after... we explore the world outside.

**Sandbox**  
units 1-4

**Real repos**  
units 5-8

**Portfolio**  
units 9-10

# How today's breakout looks (W2-W4)



# Choosing Your Breakout Room

1. We'll share the link to the entry ticket so you can indicate what unit you are on.
2. We'll create breakout groups based on the unit you selected.
3. When it's activity time, we'll share the room assignments and demonstrate how to get to your room.



## SKILL BASICS · WHAT A SKILL IS

# A skill is a folder of instructions

your unit 3 skill, installed

~/.claude/skills/plan-check/

**SKILL.md**

**Claude reads this first**

rubric.md

your checks and verdict rule

procedure.md

the steps Claude follows

references/evidence-guide.md

where the evidence lives

scope.md

which issues it may grade

voice-guide.md

your rules for comments

Plain markdown, no code. **Edit a file and the next run follows it.**

# SKILL.md says when, then how

the top of plan-check/SKILL.md

```
---
name: plan-check
description: Grade a plan package (...) and
  decide whether it is ready to post and build
  from. Use when checking a draft plan or plan
  comment before posting, ...
---

# plan-check: rubric-driven plan grading
...
Read `rubric.md`.
...
execute `procedure.md`.
```

## Always loaded

Claude knows every installed skill by these lines. The description tells it when to use this one.

## Loaded when it runs

The job, and which files to open next.

# The description picks the skill

no skill name needed: ask for the job

```
> check my plan in plan.md before I post it
```



*compared with every installed skill's description*

plan-check/SKILL.md

description: ... Use when checking a draft plan or plan comment before posting, ...



reads SKILL.md



reads rubric.md and  
procedure.md



prints a verdict

Or name it: `/plan-check` or `claude "plan-check: ..."` (Assignment step 8)

# Every skill has its own rubric

one rubric.md per skill, one question each

UNIT 1 **issue-select** rubric.md → Is this a good first issue?

UNIT 2 **repro-check** rubric.md → Is this reproduction ready to post?

**UNIT 3** **plan-check** rubric.md → Is this plan ready to post and build from?

UNIT 4 **pr-precheck** rubric.md → Is this pull request ready to submit?

You write each rubric. **The skill grades by your checks, not by Claude's own opinion.**

## UNIT 2

# Say where, not just what

one line from a unit-2 evidence guide, written twice

what the guide said

evidence to gather:

- the logs

**stuck** · which logs? where?

**add the  
where**



what Claude needed

evidence to gather:

- the logs: **run  
demo --items 6 --size 2  
copy the full output**

runs in one paste · no questions

To the author, both look done.

# The plan

Scope, diagnosis, test, comment: what a maintainer can say yes to.



## WHY PLAN FIRST

# The expensive no

feat: search commits in history tab #22653

 **Closed** nicitaacom wants to merge 5 commits into `desktop:development` from `nicitaacom:commit-search-in-history` 



github-actions Bot commented 2 weeks ago

Contributor ...

Thanks for your pull request! This is a large change (562 lines across 14 files) that doesn't reference a `help wanted` issue.

Large feature PRs require prior discussion in an issue before implementation — this helps the team assess whether the feature aligns with the project's direction before significant effort is invested.

Please **open an issue to discuss this feature first**. This PR will be automatically closed in **3 days** if requirements are not met.

► Full contribution requirements

code  
first

562 lines  
14 files · 3 days

closed

plan  
first

one comment  
one thread

read

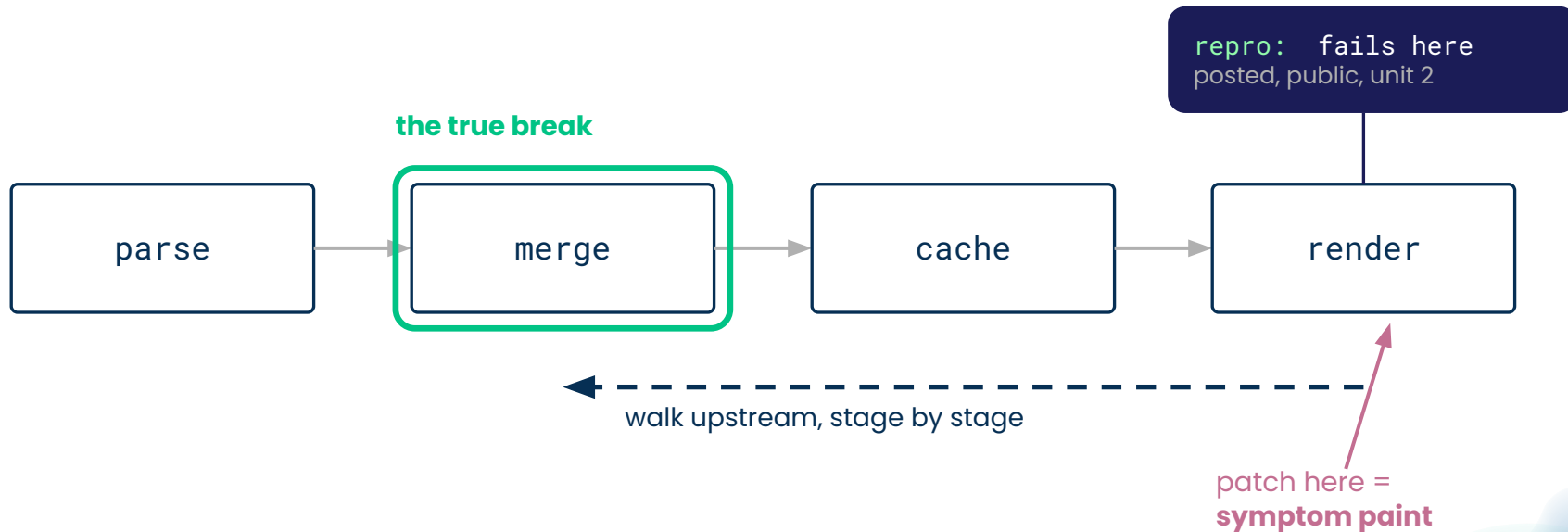
---

**Review capacity is the scarcest thing  
a maintainer owns.**

github.com/desktop/desktop · pull 22653

## ROOT CAUSE VS SYMPTOM

# Your repro is the baseline



A diagnosis must follow from the reproduced behavior. **Yours is already posted.**

# Anatomy of a plan

- |   |                  |                                                         |
|---|------------------|---------------------------------------------------------|
| 1 | <b>diagnosis</b> | floor-divide + 1 makes a ghost page                     |
| 2 | <b>scope</b>     | in: page_count() rounding<br>not in: the pagination API |
| 3 | <b>files</b>     | pager.py · test_pager.py                                |
| 4 | <b>approach</b>  | ceil-divide; drop the +1                                |
| 5 | <b>test plan</b> | re-run repro: 6 items, size 2 -> page 3 of 3            |
| 6 | <b>risks</b>     | callers that counted the ghost page                     |

**PLAN.MD**

issue #12 · you · this week

**The not-in line is the plan's boundary.**

It is what a reviewer can hold the diff to.

## SCOPE CREEP

# The plan that eats the repo

### the bounded plan

pager/

pager.py

cli.py

config.py

docs/

usage.md

tests/

test\_pager.py

**touches 2 files**

### the creep plan

pager/

pager.py

cli.py

config.py

docs/

usage.md

tests/

test\_pager.py

**touches everything it saw**

← "while I'm here: rename the helpers"

← "quick config migration"

← "docs pass, why not"

Maintainers reject this even when the code is good. **The boundary is the point.**

## THE DECISIVE TEST

# How you'll know it worked

written before you touch a line of code

PLAN.MD · PART 5 · TEST PLAN

```
run      demo --items 6 --size 2
today    4 pages
expect   3 pages (6 items, 2 per page)
```

“see if it works” is not a test plan. **A number a stranger can check is.**

**Write the expected output down first.** Then anyone can run it and watch the fix work.

## THE PLAN COMMENT

# Read the thread first

reporter  
extra page at exact multiples

maintainer signal  
keep the current API, no new flags

maintainer signal  
a repro in the thread gets reviewed first

*"Hi, I can fix this,  
please assign me."*

**HOLD**

### your comment, scaffolded

|                   |                                     |
|-------------------|-------------------------------------|
| what I found      | cause traced to page_count rounding |
| what I'll do      | fix inside the current API          |
| how I'll prove it | my posted repro, before and after   |

**Promise only what the plan contains.**

Engage what the maintainer already said.

# The build

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You drive, AI operates. No diff lands unread.



## THE OPERATING MODEL

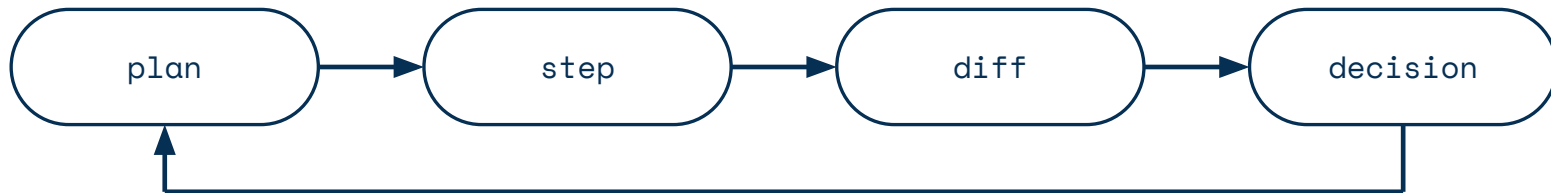
# You drive, AI operates

### THE DRIVER · you

the plan · the diff · accept / redo

### THE OPERATOR · AI

one bounded step at a time, diff back to you



No diff lands unread.

## STANDARD PRACTICE

# The industry already works this way

The last test pins the invariant that `page_count` agrees with the actual splitting done by `page()`.

### Files

- `pager.py` (fix, lines 12-14)
- `test_pager.py` (new tests)

### Verification

Run `python3 -m pytest` from `/private/tmp/w3c/planrepo`. All tests, including the existing two, should pass. `pytest 9.1.1` is already available; no install step needed (flat layout, from `pager import ... resolves from rootdir`).

---

Claude has written up a plan and is ready to execute. Would you like to proceed?

1. Yes, and use auto mode
2. Yes, manually approve edits
3. Tell Claude what to change  
shift+tab to approve with this feedback

you approve here,  
before any edit



Claude Code plan mode, captured live.

## STANDARD PRACTICE

# The industry already works this way

- Design Docs: <https://www.designdocs.dev/>
- RFCs: <https://www.ietf.org/process/rfc/>
- Plan Mode - a Claude mode where Claude reads and plans, but edits nothing until you approve at this dialog. When you run your build tonight, you ARE this dialog.  
<https://academy.claude.com/courses/ai-native-sdlc-playbook/plan-mode>

# Your tool

SKILL.md no longer lists the steps. You write them in procedure.md.



## THIS UNIT'S TOOL

# plan-check: what's fixed, what you write

the frame · fixed

modes · scope.md · output contract

judgment files · you write new ones for plans

rubric.md · references/evidence-guide.md

**procedure.md**

the skill's own operating steps

**you write the steps**

SKILL.md tells Claude to follow

**procedure.md**

## NEW THIS UNIT

# A procedure someone else can run

1

### Read order

What to read first

2

### Evidence gathering

How to gather the evidence

3

### Check execution

How to grade each check

4

### Verdict assembly

How to reach the verdict

Claude can't ask what a step means,  
**so each step has to say it.**

## THE OUTPUT CONTRACT

# The skill runs

| plan-check · pkg-07                           |      | eval mode                                                                                                |
|-----------------------------------------------|------|----------------------------------------------------------------------------------------------------------|
| bounded-scope                                 | pass | "one change, limits stated"                                                                              |
| diagnosis-grounded                            | fail | plan: "the cache returns a stale page total"<br>repro: "ran with the cache disabled; page 4 still empty" |
| executable-by-a-stranger                      | pass | "files and approach named"                                                                               |
| test-decisive                                 | pass | "expected-after stated"                                                                                  |
| comment-faithful                              | pass | "promises only what the plan holds"                                                                      |
| the verdict quotes the submission's own lines |      |                                                                                                          |



the deciding check is  
**the grounding one**

# The eval

---

Same bar, third time: 18 of 20.



## THE EVAL

# Same bar, third time: 18 of 20



■ 16 loud calls    □ 4 arguable calls

18 means you missed nothing loud and split on at most two arguable ones

**and every category scores at least once:**

clear accepts

scope creep

wrong cause

unbuildable

thread + convention

two accepts are scoped-down on purpose: deferring part of the issue, and saying so, is ready

\$0.20 per practice submission · about \$4 a full run · unlimited retries

BREAK • 5 MINUTES

# Take five

Take 5 mins to step away from the computer.  
Feel free to turn off your camera and return  
promptly after 5 mins.



THE ACTIVITY • 50 MINUTES • BREAKOUT ROOMS

# Teach Claude to Grade a Plan

## Setup • 5 min

- \* Pick a captain.
- \* The captain copies the worksheet and shares the link in the Zoom chat.
- \* Everyone works together in that one doc.

## Getting started

- \* You don't need anything to start.
- \* The worksheet gives you a **sample rubric and procedure** for Phase 1.
- \* Every step is on the Activity tab.

**BEFORE ROOMS OPEN** | Open the Unit 3 Activity tab now.

# Breakouts

## BEFORE YOU GO

# One thing, due Sunday

**Your repo link** · the same one you've submitted every unit

- \* `tools/plan-check/` with your rubric, procedure, evidence guide, and voice guide
- \* `beat-1-sandbox/unit-3/plan-and-implement.md`: plan comment, branch, test evidence, and **eval write-up**
- \* `beat-1-sandbox/unit-3/plan.md`: your plan, with Deviations filled in
- \* `beat-1-sandbox/unit-3/eval-run.txt`: one complete run, from `--save-run`

**25 PTS**

full rubric on the Grading Rubrics page  
unlimited retries before the deadline

# Submit & Land It

---

Bring your branch. It becomes your sandbox pull request, and this time you build the whole tool.

## TONIGHT

run the eval · post the plan · build · submit on the portal

## ONE GOOD THING

# Let's celebrate your wins

Reflect on today's session. Maybe you got a question answered, overcame a coding challenge, or learned how to do something completely new.

**IN THE CHAT** | share one good thing you learned or experienced today

# EARLY-COURSE SURVEY



1. **Open** the email you enrolled in CodePath with
2. **Locate** the message from [support@codepath.org](mailto:support@codepath.org) titled:  
→ **[Course Code] Early Course Survey - Please Complete**
3. **Open** your unique survey link
4. **Answer** each question completely and honestly
5. **Celebrate**: You're helping CodePath courses improve!

BEFORE YOU GO

# Final reminders

- \* Run the eval · post the plan · build · submit on the portal by 11:59pm PT on Sunday
- \* Fix-It Garages are over for the term, but you can still get setup help if needed during Study Halls, which run through the end of the term
  - \* **Cowork Cafe** happening 7–8 PM PT tonight! Catch up on your W1 – W2 assignments with folks who can help. [Zoom link](#)
- \* If you haven't yet, please fill out your Early Course survey!
- \* [LLM as a Judge](#) – Hugging Face
- \* **Next Unit:** Submit & Land It